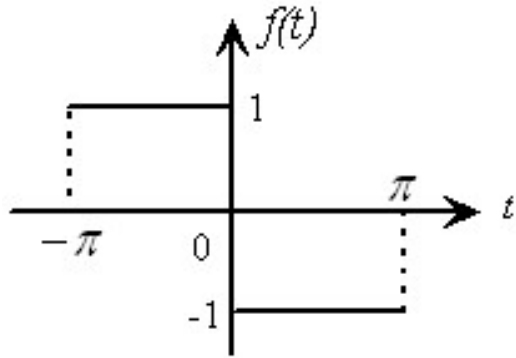




第六讲 课下作业

- 6-1 如图矩形波，请用下列函数近似，求系数



$$f(t) = C_1 \sin t + C_2 \sin 2t + \dots + C_n \sin nt$$

$$6-1 \quad f(t) = C_1 q_1(t) + C_2 q_2(t) + \dots + C_n q_n(t)$$

$$C_n = \frac{\int_{t_1}^{t_2} f(t) q_n(t) dt}{\int_{t_1}^{t_2} q_n^2(t) dt} = \frac{\int_{-\pi}^{\pi} f(t) \sin nt dt}{\int_{-\pi}^{\pi} \sin^2 nt dt}$$

$$= \frac{\int_{-\pi}^0 1 \cdot \sin nt dt + \int_0^{\pi} -1 \sin nt dt}{2 \int_0^{\pi} \sin^2 nt dt}$$

$$= \frac{-2 \int_0^{\pi} \sin nt dt}{2 \int_0^{\pi} \sin^2 nt dt}$$

$$= \frac{\frac{1}{n} \cos nt \Big|_0^{\pi}}{\frac{\pi}{2} \frac{1}{2} \Big|_0^{\pi} - \frac{1}{4n} \sin 2nt \Big|_0^{\pi}}$$

$$= \frac{2}{n\pi} [\cos n\pi - \cos(0)]$$

$$= \frac{2}{n\pi} [(-1)^n - 1], \quad n = 1, 2, 3, \dots$$

$$C_1 = -\frac{4}{\pi}, \quad C_2 = 0, \quad C_3 = -\frac{4}{3\pi}, \quad C_4 = 0, \dots, \quad C_n$$



课下作业

- 6-2 已知 $f(t) \Leftrightarrow F(j\omega)$ ，求下列信号的傅里叶变换。

(1) $tf(2t)$

(2) $(t-2)f(t)$

(3) $(t-2)f(-2t)$

(4) $t \frac{df(t)}{dt}$

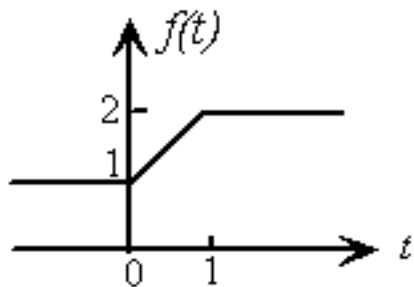
(5) $f(1-t)$

(6) $(1-t)f(1-t)$

(7) $f(2t-5)$

(8) $tU(t)$

- 6-3 求如图函数的傅里叶变换



6-2 (1) 由频域微分性质可知

$$t^n f(t) \leftrightarrow j^n F^n(\omega)$$

再由尺度变换性质可知

$$f(at) \leftrightarrow \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$$

得: $F[f(2t)] = \int_{-\infty}^{\infty} f(2t) e^{-j\omega t} dt$

$$\frac{dF[f(2t)]}{d\omega} = \int_{-\infty}^{\infty} -jt f(2t) e^{-j\omega t} dt = F[-jt f(2t)]$$

$$tf(t) \leftrightarrow jF'(j\omega)$$

$$f(2t) \leftrightarrow \frac{1}{2} F\left(\frac{j\omega}{2}\right)$$

$$\therefore tf(2t) \leftrightarrow \frac{1}{2} jF'\left(j\frac{\omega}{2}\right)$$

(2) $(t-2)f(t) = tf(t) - 2f(t)$

微分性质: $tf(t) \leftrightarrow jF'(j\omega)$

$$-2f(t) \leftrightarrow -2F(j\omega)$$

$$F[tf(t) - 2f(t)] = jF'(j\omega) - 2F(j\omega)$$

(3) $(t-2)f(-2t) = tf(-2t) - 2f(-2t)$

$$tf(-2t) \leftrightarrow \frac{1}{2} jF'\left(-j\frac{\omega}{2}\right)$$

$$-2f(-2t) \leftrightarrow -2 \cdot \frac{1}{2} F\left(-j\frac{\omega}{2}\right)$$

$$F[tf(-2t) - 2f(-2t)] = \frac{1}{2} jF'\left(j\frac{\omega}{2}\right) - F\left(j\frac{\omega}{2}\right)$$

(4) $t \frac{df(t)}{dt} = tf'(t)$

$$F\left[t \frac{df(t)}{dt}\right] = jF'\left[\frac{df(t)}{dt}\right]$$

$$F\left[\frac{df(t)}{dt}\right] = j\omega F(j\omega)$$

$$= j \frac{d[j\omega F(j\omega)]}{d\omega}$$

$$= j[jF(j\omega) + j\omega F'(j\omega)] = -F(j\omega) - \omega F'(j\omega)$$

$$(5) f(1-t)$$

$$\text{时移特性 } f(t-t_0) \leftrightarrow F(\omega) e^{-j\omega t_0}$$

$$f(1-t) \leftrightarrow F(j\omega) e^{-j\omega(1)} = F(-j\omega) e^{-j\omega}$$

$$(6) (1-t)f(1-t) = f(1-t) - tf(1-t)$$

$$F[-tf(1-t)] = j F'[-f(1-t)]$$

$$= j \frac{d[F(j\omega) e^{-j\omega}]}{d\omega}$$

$$= j [-F'(-j\omega) e^{-j\omega} - F(-j\omega) j e^{-j\omega}]$$

$$= -j F'(-j\omega) e^{-j\omega} - F(j\omega) e^{-j\omega}$$

$$F[f(1-t) - tf(1-t)] = F(-j\omega) e^{-j\omega} - F(j\omega) e^{-j\omega} - j F'(-j\omega) e^{-j\omega}$$

$$= -j F'(-j\omega) e^{-j\omega}$$

$$(7) f(2t-5) = f[2(t-\frac{5}{2})] \leftrightarrow \frac{1}{2} F(j\frac{\omega}{2}) e^{-j\omega(\frac{5}{2})}$$

$$(8) tU(t) \quad F[U(t)] = \pi\delta(\omega) + \frac{1}{j\omega}$$

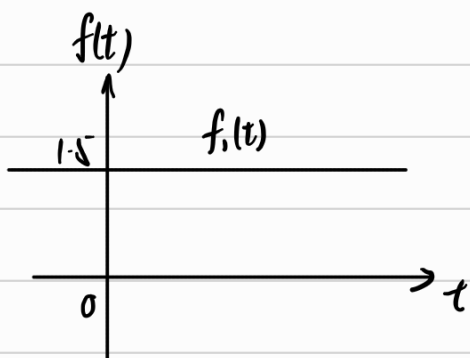
$$F[tU(t)] = j F'[U(t)]$$

$$= j [\pi\delta'(\omega) - \frac{1}{j\omega^2}]$$

$$= \pi j \delta'(\omega) - \frac{1}{\omega^2}$$

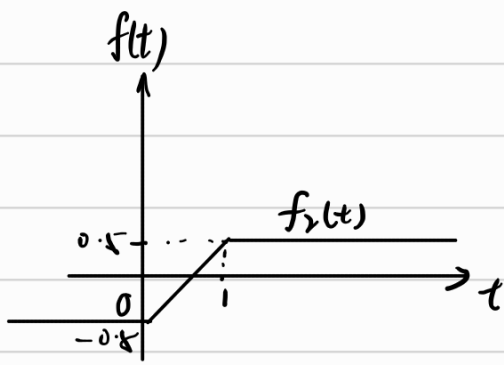
6-3

$$f(t) = f_1(t) + f_2(t)$$



$$f_1(t) = 1.5 \quad (\text{for all } t)$$

$$F_1(j\omega) = 2\pi(1.5)\delta(\omega) = 3\pi\delta(\omega)$$



$$F_2(j\omega) = \frac{1}{j\omega} \text{Sa}\left(\frac{\omega}{2}\right) e^{-j\omega\left(\frac{1}{2}\right)}$$

$$F(j\omega) = F_1(j\omega) + F_2(j\omega) = 3\pi\delta(\omega) + \frac{1}{j\omega} \text{Sa}\left(\frac{\omega}{2}\right) e^{-j\omega\left(\frac{1}{2}\right)}$$



课下作业

- 6-4 求周期余弦函数 $f(t) = E \cos(\omega_1 t)$ 的功率谱和自相关函数
- 6-5 已知 $f_1(t) \leftrightarrow E\tau \text{Sa}\left(\frac{\omega\tau}{2}\right)$, 求 $f(t) = f_1(t) * f_1(t)$ 的频谱密度函数 $F(\omega)$

6-4

$$f(t) = E \cos(\omega_1 t) = E \left(\frac{e^{j\omega_1 t} + e^{-j\omega_1 t}}{2} \right)$$

$$F[f(t)] = \frac{E}{2} 2\pi [\delta(\omega - \omega_1) + \delta(\omega + \omega_1)] \\ = E\pi [\delta(\omega - \omega_1) + \delta(\omega + \omega_1)]$$

$$S(\omega) = 2\pi \sum_{n=-\infty}^{\infty} |F_n|^2 \delta(\omega - n\omega_1)$$

$$S(\omega) = 2\pi \left[\frac{E^2}{4} \delta(\omega - \omega_1) + \frac{E^2}{4} \delta(\omega + \omega_1) \right] \\ = \frac{E^2 \pi}{2} [\delta(\omega - \omega_1) + \delta(\omega + \omega_1)]$$

$$R(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f(t) f(t+\tau) dt$$

$$f(t) = E \cos(\omega_1 t)$$

$$R(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} E \cos(\omega_1 t) \cdot E \cos[\omega_1 (t+\tau)] dt$$

$$\begin{aligned} \cos(\omega_1 t) \cdot \cos[\omega_1 (t+\tau)] &= \cos(\omega_1 t) \cdot \cos(\omega_1 t + \omega_1 \tau) \\ &= \cos(\omega_1 t) \cdot [\cos(\omega_1 t) \cos \omega_1 \tau - \sin(\omega_1 t) \sin(\omega_1 \tau)] \\ &= \cos^2(\omega_1 t) \cos(\omega_1 \tau) - \cos(\omega_1 t) \sin(\omega_1 t) \sin(\omega_1 \tau) \\ &= \left[\frac{1}{2} + \frac{1}{2} \cos(2\omega_1 t) \right] \cos(\omega_1 \tau) - \frac{1}{2} \sin(2\omega_1 t) \sin(\omega_1 \tau) \\ &= \frac{1}{2} \cos(\omega_1 \tau) + \frac{1}{2} \cos(2\omega_1 t) \cos(\omega_1 \tau) \\ &\quad - \frac{1}{2} \sin(2\omega_1 t) \sin(\omega_1 \tau) \\ &= \frac{1}{2} [\cos(\omega_1 \tau) + \cos(2\omega_1 t + \omega_1 \tau)] \end{aligned}$$

$$R(\tau) = \frac{E^2}{2} \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \cos(\omega_1 \tau) + \cos(2\omega_1 t + \omega_1 \tau) dt$$

$$\int_{-T/2}^{T/2} \cos(\omega_1 \tau) dt = \cos(\omega_1 \tau) T$$

$$\int_{-T/2}^{T/2} \cos(2\omega_1 t + \omega_1 \tau) dt = \left. \frac{\sin(2\omega_1 t + \omega_1 \tau)}{2} \right|_{-T/2}^{T/2}$$

$$= \frac{\sin(2\omega_1 \tau/2) + \omega_1 \tau}{2\omega_1} - \frac{\sin(\omega_1 \tau/2) + \omega_1 \tau}{2\omega_1}$$

当 $T \rightarrow \infty$, 以上项为 0

$$\therefore R(\tau) = \frac{E^2}{2} \cdot \frac{1}{\tau} \cdot \cos(\omega_1 \tau) \cdot \tau$$

$$R(\tau) = \frac{E^2}{2} \cos(\omega_1 \tau)$$

6-5

$$f_1(t) \leftrightarrow E\tau \text{Sa}(\frac{\omega\tau}{2})$$

$$f(t) = f_1(t) * f_1(t) \leftrightarrow F_1(\omega) \cdot F_1(\omega)$$

$$= E\tau \text{Sa}(\frac{\omega\tau}{2}) \cdot E\tau \text{Sa}(\frac{\omega\tau}{2})$$

$$F(\omega) = E^2 \tau^2 \text{Sa}^2(\frac{\omega\tau}{2})$$